For the English phrase “I ate food” in figure 10.7, we would use all three words. The first Chinese word would be produced by taking a weighted sum of all the English words in the sentence: $c_i = f(\Sigma a_i \cdot e_i)$, where a_i is the (attention) weight, which is a number between 0 and 1 such that $\Sigma a_i = 1$. The function f would be a neural network, such as a simple feedforward neural network. The function as a whole would need to learn the neural network weights in f as well as the attention weights, a_i. The attention weights would be produced by some other neural network function.

After successful training, we can inspect these attention weights and see which English words are attended to when translating to a given Chinese word. For example, when producing the Chinese word 我, the English word “I” would have a high attention weight associated with it, whereas the other words would be mostly ignored.

This general procedure is called *kernel regression*. To take an even simpler example, let’s say we have a data set that looks like figure 10.8, and we want to make a machine learning model that can take an unseen x and predict an appropriate y, given this training data. There are two broad classes of how to do this: *nonparametric* and *parametric* methods.

Figure 10.8. Scatter plot of a nonlinear data set on which we might want to train a regression algorithm.